

Phase 10A - README

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PHASE 10A — Decision Experience & Power BI (Data Marts + Dashboards 1-7)

Real Estate AI Decision Intelligence Platform

Phase 10 was split into 10A (this phase: Power BI architecture + all 7 dashboards) and 10B (the AI Explanation Layer, §10.10 — next). Matches the established convention: governed-output-only data layer, documented architecture, automated build scripts.

What was built

Data marts (`data_marts/*.csv` , 12 files) — built by `src/marts/build_marts.py` , reading exclusively from Phase 6/7/8/9's already-tested CSV outputs (no financial/risk/geo/decision math recomputed here).
`src/marts/test_phase10a.py` — 9 automated tests, all passing (added during a post-delivery audit that also caught an inherited Phase 6 ROI bug in the source CSVs – see `docs/phase10a_report.md` Section 5B).

7 dashboards (`dashboards/*.html`) — self-contained, data-accurate HTML pages:

File	Dashboard
<code>index.html</code>	1 — Executive Overview
<code>dashboard_02_market_locality.html</code>	2 — Market & Locality Intelligence
<code>dashboard_03_property_investment.html</code>	3 — Property & Investment Analysis
<code>dashboard_04_financial_risk.html</code>	4 — Financial & Risk Intelligence
<code>dashboard_05_geo.html</code>	5 — Geo Intelligence
<code>dashboard_06_decision.html</code>	6 — Decision Intelligence
<code>dashboard_07_data_quality.html</code>	7 — Data Quality & Monitoring

Open any of these directly in a browser — they're fully self-contained (Chart.js + Google Fonts load from CDN, everything else is embedded).

`docs/power_bi_architecture.md` — star-schema data model, DAX measures library, and a literal build guide for turning these marts into a real Power BI `.pbix` .

Full detail: `docs/phase10a_report.md` , `docs/limitations.md` .

Why HTML mockups instead of a .pbix file

This build environment has no Power BI Desktop and cannot produce a `.pbix` binary. Instead, this phase delivers the two things that actually determine whether a real Power BI build will be correct: clean governed data marts, and data-accurate previews of every dashboard, plus a complete build guide. Stated plainly in the report rather than glossed over.

A real bug found and fixed

Visual QA on Dashboard 3 caught `property_id=1` showing “nan” for its locality — a join-scope bug (locality name was sourced only from properties *with* full financial data, and property 1 was deliberately included without that data to demonstrate the INSUFFICIENT decision path). Fixed by sourcing locality name independently for every property. Details in `docs/phase10a_report.md` Section 5.

How to run

```
cd phase10a
pip install -r requirements-phase10a.txt --break-system-packages

python3 src/marts/build_marts.py           # rebuilds data_marts/*.csv
python3 src/dashboards/dashboard_01_executive.py
python3 src/dashboards/dashboard_02_market_locality.py
python3 src/dashboards/dashboard_03_property_investment.py
python3 src/dashboards/dashboard_04_financial_risk.py
python3 src/dashboards/dashboard_05_geo.py
python3 src/dashboards/dashboard_06_decision.py
python3 src/dashboards/dashboard_07_data_quality.py
# -> writes/refreshes dashboards/*.html
```

Then open any file in `dashboards/` in a browser.